

Project Milestone 1: Sales and Profit Analysis for Retail Decision Support Using the Global Superstore Dataset

- Sahanabanu Shaikh
- Student number - **501342966**
- Supervisor's name ("TBA")
- Date of submission – Jan 26 2026

1. Theme

The selected theme for this project is **Classification and Regression using a non-textual dataset**. This theme focuses on analyzing structured numerical and categorical data to extract insights and build predictive models. Classification and regression techniques are widely used in business analytics to support decision-making such as pricing, discount planning, and inventory management. This project applies these techniques to retail sales data to understand factors affecting profit and to support data-driven business decisions

2. Business Problem & Motivation

Retail businesses must regularly make decisions about pricing, discounting, and inventory planning. Offering discounts can increase sales volume, but excessive or poorly planned discounts may reduce overall profit. Managers need data-driven insights to understand which products and discount levels lead to profitable outcomes.

The motivation of this project is to analyze historical retail sales data in order to identify the key factors that influence profit and to develop predictive models that can support better pricing and discount decisions. By learning from past transactions, businesses can reduce the risk of loss-making sales and improve profitability.

3. The project aims to answer the following research questions

RQ-1 Which product categories, sub-categories, discounts, and quantities are most strongly associated with higher profit?

RQ-2 How accurately can future profit be predicted from historical sales data such as discount, quantity, category, and region?

RQ-3 Can sales transactions be classified as profitable or non-profitable in advance to support discount and pricing decisions?

4. Retail managers must decide which products to discount, how much discount to offer, and which items to prioritize for inventory. Poor discounting decisions can increase sales volume but reduce profit.

This project uses historical sales data to support pricing and discount decisions by identifying which factors most strongly influence profit and by predicting whether a sale will be profitable.

5. Dataset Description

Dataset: Global Superstore Dataset (Kaggle)

Granularity:

Each row represents a single sales transaction (one product sold in one order).

Time span:

The dataset covers multiple years of historical retail sales data.

Size:

Approximately 50,000 records with both numerical and categorical variables.

Key variables:

- Sales (continuous)
- Profit (continuous)
- Quantity (numeric)
- Discount (numeric)
- Category and Sub-Category (categorical)
- Region and Segment (categorical)

Target definitions:

- **Regression target:** Profit
- **Classification target:** Profitability label (Profitable = Profit > 0, Not Profitable = Profit ≤ 0)

Limitations:

- The dataset does not include marketing spend or customer demographics
- Results may not generalize beyond retail businesses
- Profitability is influenced by external factors not captured in the dataset

6. Evaluation setup:

The dataset will be split into training (80%) and testing (20%) sets.

Regression evaluation:

- Baseline model: Predicting the mean profit for all transactions
- Models will be evaluated using RMSE and R^2 score

Classification evaluation:

- Labeling rule:
 - Profitable = Profit > 0
 - Not Profitable = Profit ≤ 0
- Baseline model: Always predicting the majority class
- Evaluation metrics: Accuracy, Precision, Recall, and F1-score

7. The project will be completed using the following steps:

1. Data Preparation
 - Cleaning the data
 - Handling missing values
 - Converting categorical variables into numerical form
2. Exploratory Data Analysis (EDA)
 - Summary statistics (mean, median, variance)
 - Basic charts to understand trends and relationships
3. Model Building
 - Applying regression models such as Linear Regression and Decision Tree Regression
 - Optional classification to group profitable and non-profitable sales
4. Model Evaluation
 - Metrics for regression: Root Mean Squared Error (RMSE), R² Score
 - Metrics for classification: Accuracy, Precision, Recall, F1-score
5. Tools
 - Python programming language
 - Libraries: Pandas, NumPy, Scikit-learn, Matplotlib

8. References

1. Fatih Ilhan. (n.d.). *Global Superstore Dataset*. Kaggle. Retrieved from <https://www.kaggle.com/datasets/fatihilhan/global-superstore-dataset>
2. Géron, A. (2019). *Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow* (2nd ed.). O'Reilly Media.
3. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning: With Applications in R* (2nd ed.). Springer.
4. Raschka, S., & Mirjalili, V. (2022). *Python Machine Learning* (4th ed.). Packt Publishing.
5. Han, J., Kamber, M., & Pei, J. (2012). *Data Mining: Concepts and Techniques* (3rd ed.). Morgan Kaufmann.
6. Scikit-learn Developers. (2025). *Scikit-learn: Machine Learning in Python*. Retrieved from <https://scikit-learn.org>